

Hana Aliyah Mufidah

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PROFESSIONAL SUMMARY

Backend and full-stack engineer with **6+ years** building scalable, reliable systems for B2B payments and work-management platforms. Designed and operate a **multi-chain block indexer** built from coordinated workers: a forward indexer for real-time block scanning, a configurable backward indexer for gap backfill and outage recovery, and periodic cleanup jobs that keep the database lean, with Redis dedup so each transaction is processed once. Comfortable **owning systems end to end**: asynchronous processing, worker design, data integrity, and secure delivery on AWS with CI/CD.

CORE COMPETENCIES

Coordinated Worker Design

Asynchronous / Event Pipelines

Outage Recovery & Backfill

Backend Architecture

Idempotency & Deduplication

AWS & CI/CD

PostgreSQL / Redis

RPC & API Integration

WORK EXPERIENCE

Nespay

Apr 2025 - Present

Software Developer - Backend & Full-Stack

- Architected and built a **multi-chain block indexer from coordinated workers**: a forward indexer for real-time block scanning, a configurable backward indexer for gap backfill and outage recovery, and periodic cleanup jobs to prune stale data and keep PostgreSQL lean, with **Redis dedup so each transaction is processed once**. It relays on-chain payment data to backend services.
- Developed **core application architecture**; set up AWS authorization and CI/CD pipelines (GitHub Actions on EC2).
- Delivered **wallet transactions, invoice management, and external wallet integrations** (Xellar, Privy); integrated RPC services and KYC compliance (Sumsu, Kredibel).
- Led the **Backoffice Dashboard (React.js/Next.js)** and backend server for user and admin interfaces; ran peer code reviews for scalable, high-quality code.

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Feb 2020 - Dec 2024

Full Stack Software Engineer

- Built an **idempotent integration service** to sync data from an HRIS into the platform, automating a fragile manual reconciliation and preventing duplicate or corrupted records.
- Enhanced **praise/endorsement features**: department/division filters, leaderboards, and skill-matched people search.
- Improved the employee time-off system and work timeline; simplified bulk approvals; integrated time-off with public holidays and objectives.
- Maintained a legacy codebase; stabilized input-component behavior and validation across the application.

Ultra Voucher

Nov 2018 - Jan 2020

Full Stack Web Developer

- Built a **web app to generate digital voucher codes** for affiliate providers, tracking status and generating reports, replacing a manual, untracked Excel-based process.
- Developed various dashboard applications for business needs.

PROJECTS

Multi-Chain Block Indexer Systems / Reliability

A relay that **watches the chain and notifies backend services when a transaction has been paid**. Redesigned from a single loop into coordinated workers: a forward indexer for real-time scanning, a configurable backward indexer that backfills gaps and can re-scan after an outage, and periodic cleanup to avoid database bloat. Redis dedup handles the overlapping re-scans so nothing is processed twice.

Node.js · PostgreSQL · Redis · RPC · webhooks · multi-chain

EDU Chain Hackathon · DeFi Track 3rd Place

Placed 3rd in the DeFi track (May 2025), building on the wallet, on/off-ramp, and RPC integration work shipped at Nespay.

EDUCATION

B.S. Geophysical Engineering, Institut Teknologi Bandung	2015
JavaScript Fullstack Immersive Program, Hacktiv8	2018

CERTIFICATIONS

AWS Certified Cloud Practitioner · Amazon Web Services	2024
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SKILLS

Backend: Node.js, Nest.js, distributed workers, async/event pipelines, RPC, REST APIs, webhooks **Languages:** TypeScript, JavaScript, Go (basic)
Frontend: React.js, Next.js **Data:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Neo4j
Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, GitHub Actions CI/CD, Ansible, Vagrant **Testing:** Nightwatch, Selenium, Mocha
Web3: wallet integrations (Xellar, Privy), crypto-to-fiat on/off-ramp, KYC (Sumsu) **Spoken:** English (Professional), Indonesian (Native), Korean (Conversational)